

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 20

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Stress-related illness: ulcers, high blood pressure, heart disease (1) • Behavioural changes: irritability, aggression, lack of concentration leading to accidents (1) • Communication problems / educational underperformance (eg schools under flight paths) (1) • Sleep disturbance (1)	3
01.2	<i>1 mark for each point.</i> • Acoustic fatigue: objects exposed to sound at their natural resonant frequency vibrate, causing stress cracking / structural failure (eg early jet engine cowlings) (1) • Vibration damage: repetitive vibration from heavy vehicles damages buildings, bridges, underground pipes (1) • Shock impacts: the force of a sudden very loud noise (sonic boom) breaks windows / dislodges masonry (1)	3
01.3	<i>1 mark for each point.</i> • Birds that sing to establish territories / attract mates cannot be heard against background noise (1) • Birds scared by sudden noise leave nests so predators take eggs / chicks; cliff-nesting seabirds knock eggs or chicks into the sea (1)	2
01.4	<i>1 mark for each point.</i> • Whales / dolphins use hearing / echolocation to find food and communicate; loud sonar causes hearing damage if close to the source (1) • Behavioural changes / disorientation possibly linked to mass strandings; reduced feeding success (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 287; AQA spec 3.4.3.2.13.	

Question	Marking guidance	Marks
02.1	<i>Correct answer only.</i> • B - Steam washing (hot water / steam jets kill organisms in the sediment) (1)	1
02.2	<i>1 mark for each point.</i> • Chemicals cause the oil molecules to join together / solidify (1) • Producing more solid material that can be collected / lifted more easily than liquid oil (1)	2
02.3	<i>1 mark for each point.</i> • Air contains about 210 cm³ of oxygen per litre whereas water at 20 °C holds a maximum of about 6 cm³ per litre (less than 3% of the amount in air) (1) • So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen; many aquatic organisms are near the lower end of their range of tolerance (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	
03.1	<i>1 mark for each point.</i> • CO₂ rises from about 412 ppm in January to a peak of 416 ppm in May, falls to a minimum of 407 ppm in September, then rises again (1) • During spring / summer, photosynthesis by plants (growing season) removes CO₂ faster than respiration releases it, so the concentration falls (1) • In autumn / winter photosynthesis is reduced (leaf fall) but respiration and decomposition continue, so CO₂ rises (1) • The atmosphere is in dynamic equilibrium: the concentration fluctuates around a mean; more land / vegetation in the northern hemisphere makes the cycle large (1)	4
03.2	<i>1 mark for each point.</i> • Natural processes that add gases (respiration, combustion, volcanoes, denitrification) are balanced by processes that remove them (photosynthesis, dissolving, nitrogen fixation) (1) • So the average composition stays the same over long timescales although the rates vary and concentrations fluctuate around a mean (1)	2
03.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Absorbs harmful UV / solar wind radiation in the stratosphere (ozone layer) (1) • Greenhouse gases delay the escape of infrared, keeping the surface warm (33 °C warmer than without) (1) • Atmospheric pressure allows liquid water to exist (1) • Distributes heat from the tropics to higher latitudes by winds (1) • Transports water vapour to areas that would otherwise be dry (1)	2
03.4	<i>1 mark for each point.</i> • Troposphere (1) • Stratosphere (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89; AQA spec 3.2.1.1.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> • Renewable: re-forms naturally relatively quickly so use does not necessarily reduce future availability (solar, wind, tidal, geothermal, biofuel) (1) • Depletable: use can reduce future availability - all non-renewables (fossil fuels, uranium) plus renewables exploited unsustainably (1) • Wood / biofuels are renewable but depletable if forests are felled faster than they regrow (1)	3
04.2	<i>1 mark for each point.</i> • Other factors (technology, location, cost) restrict how much can be harnessed (1) • Examples: coal too deep to mine; winds high above the ground / far offshore where turbines cannot be placed; waves in deep water where equipment cannot be anchored; low-intensity sunlight that cannot produce high temperatures (1) • New technologies or higher prices may make these usable in future (1)	3
04.3	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Wind: strong reliable winds - shallow seas, open plains, uplands; low land-use conflict (not near urban areas, ecologically sensitive or scenic areas); near the grid (1) • HEP: high reliable rainfall, large catchment, narrow valley for a small dam with a large basin, impermeable stable bedrock (1) • Solar: high light intensity, low cloud cover (1) • Wave: coasts with strong reliable prevailing winds and a long fetch (1) • Tidal: large tidal range, coastal features (estuaries, channels) that focus the flow (1) • Geothermal: hot rocks near the surface / recent volcanic activity (1) • Biofuels: nearby farmland / forest / urban waste to avoid transporting bulky low-density fuel (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-177; AQA spec 3.3.2.	
05.1	<i>1 mark for each point.</i> • Photons of light absorbed by the (n-type silicon) upper layer dislodge electrons, making it negatively charged (1) • Electrons flow through an external conductor / circuit to the relatively positive lower layer; the current powers appliances (1)	2
05.2	<i>1 mark for each point.</i> • Smooth cells reflect about 30% of the light; grooved / textured surfaces (mimicking moth eyes) reflect light into the cell rather than away (1) • Multi-junction cells have several layers of different materials, each absorbing different wavelengths (1) • So a greater proportion of the available light is absorbed and converted to electricity (1)	3
05.3	<i>1 mark for each point.</i> • Solar farm: polycrystalline / thin film - lowest cost per unit electricity where land area is not limiting (efficiency per m² matters less) (1) • Satellite: multi-junction - highest efficiency where mass / area is critical and cost is less important (1) • Cost per watt = cost / efficiency: thin film 0.125, poly 0.125, mono 0.136, multi-junction 0.625 per unit (1)	3

Question	Marking guidance	Marks
05.4	<i>1 mark for each point.</i> - Intermittent: no output at night and low in winter (daily and seasonal cycles) - predictable (1) - Unpredictable: cloud cover, dust and smoke reduce intensity and scatter light and cannot be forecast accurately (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.	
06.1	<i>1 mark for each point.</i> - Numbers fell from about 210 pairs in 1945 to a minimum of about 25 in 1965 (steepest fall 1955-1960) (1) - Numbers then recovered / rose steadily after 1970, reaching about 190 by 1985 (not yet back to 1945 level) (1)	2
06.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> $(200 - 25) / 200 \times 100$ (1) $= 87.5\%$ (1)	2
06.3	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> - DDT is persistent (remains for decades) and liposoluble so it bioaccumulates in prey (small birds) and biomagnifies in sparrowhawks (top predator) (1) - Sub-lethal doses caused thin eggshells / reduced breeding success so the population declined even without adult deaths (1) - Higher doses caused sterility / death (1) - After restrictions, environmental concentrations slowly fell (persistence explains the delay before recovery) (1) - Breeding success recovered so the population rose (1)	4
06.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - DDT persists in soil / sediments / food chains for many years so exposure continued (1) - Birds of prey are k-selected: few young per year, slow to mature so populations grow slowly (1) - Other pressures (habitat loss, persecution) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
07.1	<i>1 mark for each point.</i> 2020 has higher peak flow in the wet season (32 vs $22 \text{ m}^3 \text{ s}^{-1}$ in March) (1) 2020 has much lower dry-season flow (2 vs $9 \text{ m}^3 \text{ s}^{-1}$ in July / August) (1) - Flow is much more variable / seasonal in 2020; the river almost dries up (1)	3
07.2	<i>1 mark for each point.</i> - In 1990 snow / ice stored wet-season precipitation and released it gradually as meltwater during the dry season, evening out the flow (1) - With less ice, precipitation now falls as rain / melts immediately and runs off quickly, increasing wet-season flow and flood risk (1) - Little stored ice remains to melt in the dry season so flow falls very low (1)	3
07.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Water shortages for irrigation / drinking / HEP in the dry season (1) - Flooding in the wet season (1) - Sudden floods from ice-lake outbursts (glacial lake outburst floods) (1)	2

Question	Marking guidance	Marks
07.4	<i>1 mark for each point.</i> • Surface meltwater flows down cracks to the base, lubricating the ice so it slides faster; if this exceeds the faster melting at the front, the front advances (1) • But if snowfall at the source has not increased, total volume falls; both retreating and extending glaciers can indicate warming (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.	
08.1	<i>1 mark for each point.</i> • t-test (1) • The data are measured (continuous) values and two means are being compared (1) • The data are (assumed to be) normally distributed (1)	3
08.2	<i>1 mark.</i> • Use the Mann-Whitney U test instead (compares medians) (1)	1
08.3	<i>1 mark for each point.</i> • There is a 2% probability that the difference occurred by random chance (1) • As this is less than 0.05 the difference is statistically significant / the null hypothesis is rejected (1)	2
08.4	<i>1 mark for each point.</i> • The ranges of values (mean plus or minus SD) do not overlap: woodland up to 1.6 and clearing down to 2.4 (1) • About 68% of values lie within one SD of each mean so the two data sets are clearly separated (1)	2
08.5	<i>1 mark for each point.</i> • Significance shows the difference is unlikely to be due to chance, not that a causal relationship exists (1) • Another variable (eg topography / aspect of the sites) could cause both; a controlled experiment or further evidence is needed (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
09.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Volume of ash is much less than the original waste (less landfill) (1) • Heat used for district heating / electricity generation (energy recovery) (1) • No sorting / complicated management needed (1) • Destroys pathogens / flammable organic wastes converted to CO₂ and water (1) • Refuse-derived fuel can replace fossil fuels (eg cement kilns) (1)	3
09.2	<i>1 mark for each point.</i> • High temperature and rapid cooling reduce the formation of toxic dioxins (from organic wastes reacting with chlorine) (1) • Scrubber dissolves hydrogen chloride (from chlorine in PVC) which is then neutralised with alkali / limestone; also removes heavy metals for storage at high pH (1) • Wet waste needs extra fuel to burn and lowers combustion temperature; composting recovers its nutrient value (1) • Recycling conserves the resource value of metals, glass and plastics that would otherwise be lost (1)	4

Question	Marking guidance	Marks
09.3	1 mark for each point. - Ash may contain concentrated heavy metals / toxic residues (1) - Must be stored in solid form at high pH / hazardous waste landfill so the metals remain insoluble (1)	2
09.4	Any 1 from the points below - 1 mark each, maximum 1. - Fuel (natural gas) needed to burn wet / non-flammable wastes (1) - Cost of treating effluent gases (SO₂, NO_x, HCl, smoke) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 283-284; AQA spec 3.4.3.2.12.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - State of matter: solids deposit near the source, gases disperse widely in the atmosphere; density (lead dust, hydrogen cyanide at Bhopal) controls how far particles/gases travel - Persistence / half-life: persistent pollutants (organochlorines, CFCs, heavy metals, radioactive wastes with long half-lives) travel further, accumulate and need containment rather than dilution; non-persistent pollutants (sewage, organophosphates) can be allowed to degrade - Toxicity, specificity, mode of action (enzyme inhibition, neurotoxins) determine acceptable concentrations and which species are at risk; LD50 / MDAF / MDNF used to set doses - Solubility in water (nitrates disperse and leach - control by nitrate vulnerable zones) vs liposolubility (DDT, mercury bioaccumulate and biomagnify - control by bans / restrictions) - Reactivity: reactive pollutants may degrade quickly or form secondary pollutants (ozone, acid rain, PANs) - controlling primary pollutants (catalytic converters, FGD) prevents secondary pollutants - Adsorption onto clay / sediments immobilises metals and caesium; can be used deliberately (activated carbon filters, polymer adsorption) - Mutagenic / carcinogenic / teratogenic action means no safe dose; exposure kept As Low As Reasonably Achievable (ALARA) - Synergism (ozone + SO₂; smoke + SO₂) means pollutants must be considered together - Environmental features (temperature inversions, currents, pH, oxygen) interact with properties: Critical Pathway Analysis uses properties + environment to predict where pollutants go and to focus monitoring - Evaluation: properties of new chemicals can be predicted from similar ones (precautionary principle); understanding allows prevention rather than clean-up	9

Question	Marking guidance	Marks
10.2	1 mark. The increase in the amount / concentration of a substance within an (individual) organism over time, because intake exceeds excretion or breakdown (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	
11.1	Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) - Scale of the problem: how energy is used (transport, buildings, industry, water); links between affluence and per capita use - Transport: bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles - Buildings: orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour - Industry: heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management - Limits: rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour - Evaluation with quantitative examples; comparison with the cost of new supply; conclusion that conservation is the cheapest first step but cannot alone meet demand growth Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	25

Question	Marking guidance	Marks
11.2	Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Composition and structure: N₂, O₂, Ar, CO₂, ozone; troposphere and stratosphere; pressure and liquid water; gases for biological molecules and industry • Energy processes: UV absorbed by O / O₂ / O₃ dynamic equilibrium; visible light absorbed at the surface; infrared absorbed by greenhouse gases (33 °C warming); heat distribution by winds and ocean currents; transport of water vapour; energy budget and albedo • Global climate change: anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species; difficulties of monitoring and prediction; control strategies and agreements • Ozone depletion: CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • Other threats: aerosols, smoke, land-use change; interconnection of processes • Evaluation: which functions are most at risk; success of ozone controls vs slow progress on greenhouse gases; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	25